

# How Far Does Audio Help Meeting Summarisation? The Limited Role of Prosody in Speech LLMs

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**Abstract.** Speech large language models can summarise meetings directly from audio, often assumed to beat the transcript because audio preserves prosody: the pitch, loudness, and timing that transcription discards. We show that the audio advantage is narrow and bounded, and identify why. On Voxtral-Small-24B, a duration-matched comparison on AMI shows audio matching or exceeding text in semantic similarity and in lexical overlap on commitment-bearing sections. On ICSI, the semantic edge narrows to parity and the lexical advantage reverses toward text. Two modality-blind LLM judges show no overall winner, as text covers more content while audio’s only consistent edge, fidelity per asserted decision, holds on AMI. Testing pitch, energy, and pause as mechanisms, linear probing shows that this detail is largely discarded at the multimodal adapter, and ablation shows that removing any of them leaves quality unchanged. In this model and setting, the advantage is thus capped by adapter-level attenuation and lexical redundancy.

**Keywords:** audio, speech LLMs, meeting summarisation, prosody, probing

## 1 Introduction

In spoken communication, meaning is conveyed through more than words. In addition to lexical content, speakers convey information through prosody, pitch, energy, and timing of the voice, which marks emphasis, signals commitment, and separates a firm decision from a tentative one [3]. In a meeting, these cues help to distinguish which proposals were adopted, which were set aside, and which questions were left open. Meeting summarisation, the task of compressing a long spoken record into a short account of decisions, actions, and open problems, is therefore a natural setting in which prosodic information might matter [13].

Many speech summarisation systems follow a cascaded pipeline in which automatic speech recognition first converts the audio into a transcript, and then

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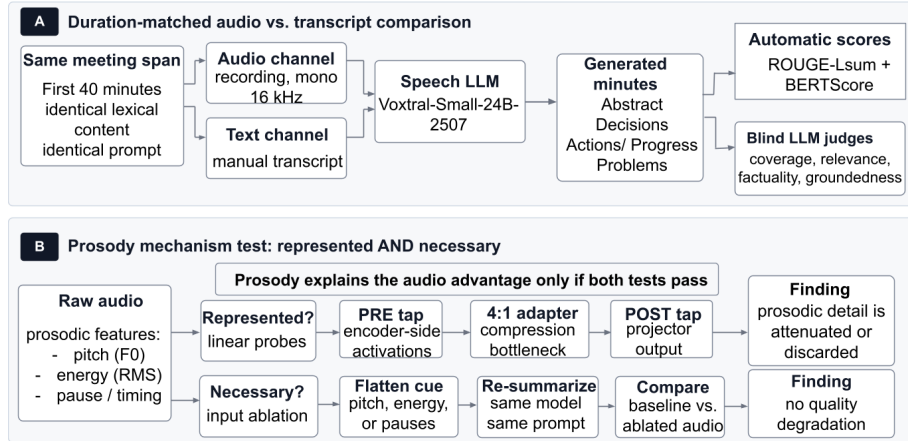
a text summarisation model summarises that transcript; this design largely removes access to speech-specific cues such as prosody, intonation, rhythm, and emphasis [14]. Speech large language models, which consume audio directly, have recently made it possible to bypass this step, and are frequently motivated by the argument that audio preserves the paralinguistic information removed by a transcript [1,2,6,16]. The expectation that follows is twofold: a model given audio should summarise a meeting better than the same model given its transcript, and prosody should be the reason.

Whether this expectation holds is not obvious, and prior evidence cuts both ways. On the one hand, prosody demonstrably carries decision-relevant cues that text lacks. However, recent work shows that much prosodic information is statistically redundant with the words themselves [19], and that the acoustic detail entering a speech LLM must survive a multimodal adapter that was trained to align speech with text, not to preserve fine prosodic structure. An advantage might therefore fail to appear, either because a strong language-model decoder already recovers the needed information from the words, or because the architecture discards the prosodic signal before it can be used. The question is thus not simply whether audio helps, but how far it helps and why.

To our knowledge, no prior study has jointly evaluated performance differences, internal representation, and causal necessity of prosodic features within a modern speech LLM. We address this using Voxtral-Small-24B-2507, an audio-native speech LLM, within the controlled evaluation framework shown in Fig. 1. We compare audio and transcript inputs under a duration-matched 40-minute cap ensuring that both conditions contain identical lexical content, so any difference is attributable to the channel rather than coverage. We complement the automatic comparison with two modality-blind LLM judges. To test whether prosody is the mechanism behind any advantage that exists, we require it to be both *represented* in the model (linear probing across the multimodal adapter) and *necessary* for the summary (input ablation of pitch, energy, and pause).

This study addresses three research questions. **RQ1:** Does direct audio input improve meeting summarisation over a duration-matched transcript, and if so, on which sections and corpora? **RQ2:** Where across the speech-to-language interface is prosodic information (pitch, energy, pause) retained or lost? **RQ3:** Are these prosodic cues necessary for the summary, such that they explain any observed audio advantage?

Across these questions, we find that the audio advantage is bounded and task-dependent rather than broad, and that the cues we test (pitch, energy, and pause) do not explain it here. We attribute the bound to two factors the analysis makes explicit: an architectural bottleneck that removes prosody at the adapter, and the redundancy of prosody with lexical content once a strong decoder is in place. These findings suggest that direct audio processing may only justify its additional computational cost when fidelity to meeting decisions is prioritised over broad content coverage.



**Fig. 1.** Controlled (A) audio-text comparison and (B) prosody mechanism test.

## 2 Related Work

*Prosody in extractive meeting summarisation.* Early meeting summarisation established that prosodic features aid extractive summarisation in feature-engineered systems, in which prosody was an explicit input for selecting salient utterances [9]. F0, energy, and duration improve utterance selection and combine additively with lexical features on ICSI, robustly across human and ASR transcripts [20]. The turn-taking and involvement features predict where the summary-worthy content lies in AMI and ICSI and multimodal fusion yields further gains [5,11]. This motivates the expectation that prosody helps. Three differences explain why that benefit need not transfer to an abstractive speech LLM. The task is generative rather than selective, so there is no utterance-selection stage for prosody to inform. Prosody must survive an encoder and a multimodal adapter rather than being supplied explicitly. The lexical channel is now a 24B language model, for which prosodic information is largely redundant with the words.

*Neural and end-to-end speech summarisation.* Direct speech-to-summary models remove the transcription step. ESSumm summarises untranscribed meetings competitively with transcript-based pipelines [18], and instruction-tuned LLMs coupled to an audio encoder summarise spoken input directly, exceeding an ASR→LLM cascade [4]. These works establish that the audio path can match or exceed the text path, but treat the advantage as a single scalar; instead we characterise its size and generality and ask what in the audio produces it. Spoken DialogSum [7] reports that an audio LLM exceeds a cascade by 28% on *emotion*-focused summaries of expressively synthesized speech, which delimits rather than contradicts our scope: factual, commitment-bearing (Decisions, Actions, Problems) minutes from naturalistic speech. Existing studies primarily

evaluate end-task performance, whereas our work investigates the mechanism underlying any observed audio advantage.

*Prosody-text redundancy and prosody in speech LLMs.* Information-theoretic analysis finds prosody to be largely redundant with text [19]. A study finds that prosody adds little when the lexical channel is clean and more when it is degraded [10]; and analysis of speech LLMs argues that standard models do not develop genuine prosody processing from pretraining alone [12]. That work diagnoses the limitation at the level of tokenisation and demonstrates recoverable prosody behaviourally on a model purpose-built to retain it, through generation and probability-based tasks such as contrastive focus, emphasis, and emotion. We instead provide mechanistic evidence on a deployed, unmodified speech LLM, using linear probing to locate where prosodic detail is lost across the adapter and input ablation to test whether it is necessary for the summary.

### 3 Experimental Setup

#### 3.1 Models, data, and conditions

The model was chosen by a screening experiment over three publicly available HuggingFace speech LLMs spanning an order of magnitude in scale and two families: Voxtral-Mini-3B-2507<sup>1</sup>, Phi-4-multimodal-instruct<sup>2</sup>, and Voxtral-Small-24B-2507<sup>3</sup>. Each model summarised the 20-meeting AMI test set under the identical protocol (same prompt, greedy decoding, per-section scoring), and we compared audio against text by the audio–text difference in ROUGE-Lsum and BERTScore. Only Voxtral-Small-24B showed a positive difference in both metrics (ROUGE-Lsum +0.008, BERTScore +0.009), increasing to +0.020 and +0.012 in the commitment-bearing sections, whereas Voxtral-Mini-3B was at parity and Phi-4-mm favoured text on lexical overlap (-0.042). We carried the same three-model comparison to the six-meeting ICSI test set, where no model showed an audio advantage on either metric, consistent with the corpus-specific effect we report in Section 4.1. We therefore study Voxtral-Small-24B, the only model with an audio advantage to explain, evaluating it on both AMI and ICSI; the study is thus explanatory of that advantage rather than representative of speech LLMs in general.

We evaluate the two corpora of meetings with manually annotated abstractive summaries, AMI<sup>4</sup> and ICSI<sup>5</sup>. For AMI we use the official *scenario-only* partition; the held-out test set comprises 20 meetings from five series (ES2004, ES2014, IS1009, TS3003, TS3007), with development (ES2003, ES2011, IS1008, TS3004, TS3006) and training sets defined accordingly. We use the Mix-Headset

<sup>1</sup> <https://huggingface.co/mistralai/Voxtral-Mini-3B-2507>

<sup>2</sup> <https://huggingface.co/microsoft/Phi-4-multimodal-instruct>

<sup>3</sup> <https://huggingface.co/mistralai/Voxtral-Small-24B-2507>

<sup>4</sup> <https://groups.inf.ed.ac.uk/ami/corpus/datasets.shtml>

<sup>5</sup> <https://groups.inf.ed.ac.uk/ami/icsi/download/>

channel, manual transcripts, and gold summaries organised into the sections such as Abstract, Decisions, Actions, and Problems. ICSI consists of naturally occurring research meetings and uses the same scheme, except that Progress replaces Actions and we adopt the six-meeting test set standard in the ICSI summarisation literature [15] (Bed004, Bed009, Bed016, Bmr005, Bmr019, Bro018).

All experiments run Voxtral-Small-24B in bfloat16 on a single NVIDIA A100 80GB GPU with greedy decoding (`do_sample=False`, `max_new_tokens=1024`). Each meeting is summarised under two conditions aligned to the same first 40 minutes (model’s processing limit) and differing only in channel: audio (the recording, mono 16 kHz) and text (the human transcript of the same span). Both conditions use the same model, Voxtral-Small-24B, and the text condition receives the manual reference transcript, so the comparison is within-model and the text condition is an error-free upper bound rather than an ASR cascade. Both carry identical lexical content over the same span, so any difference is attributable to the channel rather than coverage; truncation binds to only 6 of the 20 AMI meetings, but to all 6 ICSI meetings. The average meeting length of 20 AMI meetings is 35.31 min and 6 ICSI meetings is 57.62 min. Both conditions use the same structured prompt, instructing the model to summarise into four labelled sections. The fixed format aligns each generated section with its gold counterpart, so the audio-text comparison is scored per section.

### 3.2 Prosody Mechanism Test

We test whether these three cues are the mechanism under a fixed protocol: each must be both *represented* in the model (probing) and *necessary* for the summary (ablation). We study three measurable prosodic features whose decision relevance is established in extractive meeting summarisation [20]: *pitch* (F0), *energy* (short-time loudness, RMS), and *pause* (silence and timing).

*Probing* Linear probes were chosen because they provide a conservative estimate of linearly accessible information while maintaining interpretability. A linear probe recovers a target from frozen activations; success indicates that the feature is linearly present, while failure implies that it is not linearly recoverable there. We tap two points: *PRE-adapter*, the encoder-side layer-norm output (1280-d), and *POST-adapter*, the projector output after the 4:1 adapter (5120-d). Per-frame targets are extracted from 16 kHz audio at 50 fps: F0 via Praat (floor 75 Hz, ceiling 600 Hz, voiced frames only); energy as log-RMS; and pause as a binary low-energy-and-unvoiced label. The energy and pitch probes are ridge regressions scored by  $R^2$ , with the penalty selected on a development set; the pause probe is logistic, scored by AUROC. Probes are fit on training frames (capped at 400k) and evaluated per-meeting with group-disjoint splits, so that no speaker or meeting group is shared between fitting and evaluation; group-disjoint evaluation is essential, since within-meeting splits inflate the pitch result. The procedure (Algorithm 1) is instantiated twice: within-corpus, with  $\mathcal{T}$ ,  $\mathcal{D}$ ,  $\mathcal{E}$  the AMI training, development, and test partitions; and cross-corpus, training and tuning on AMI ( $\mathcal{T}$ ,  $\mathcal{D}$ ) and evaluating the six ICSI test meetings ( $\mathcal{E}$ ). The

**Algorithm 1** Prosody Probing Across the Adapter

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**Input:** Frozen speech LLM  $M$ ; train set  $\mathcal{T}$ , dev set  $\mathcal{D}$ , eval set  $\mathcal{E}$ ; taps {PRE, POST}; targets {pitch, energy, pause}; frame cap  $C$

- 1: **for** each meeting  $m \in \mathcal{T} \cup \mathcal{D} \cup \mathcal{E}$  **do**
- 2:    $A_m^{\text{PRE}}, A_m^{\text{POST}} \leftarrow M(\text{audio}_m)$  {forward-hook taps}
- 3:   extract  $y_m^{\text{pitch}}, y_m^{\text{energy}}, y_m^{\text{pause}}$ ; align to each tap
- 4: **end for**
- 5: **for** each target  $t$  and tap  $r \in \{\text{PRE}, \text{POST}\}$  **do**
- 6:    $X_{tr}, y_{tr} \leftarrow$  frames from  $\mathcal{T}$ , subsampled to  $C$  {voiced only for pitch}
- 7:   **if**  $t = \text{pause}$  **then**
- 8:      $P_{t,r} \leftarrow$  logistic probe on  $(X_{tr}, y_{tr})$
- 9:   **else**
- 10:      $\alpha^* \leftarrow \arg \max_{\alpha} R^2$  on  $\mathcal{D}$  {ridge penalty via dev}
- 11:      $P_{t,r} \leftarrow$  ridge probe( $\alpha^*$ ) on  $(X_{tr}, y_{tr})$
- 12:   **end if**
- 13: **end for**
- 14: **for** each target  $t$ , tap  $r$ , meeting  $m \in \mathcal{E}$  **do**
- 15:    $s_{t,r,m} \leftarrow R^2$  (pitch, energy) or AUROC (pause) on  $m$
- 16: **end for**
- 17: **return** per-tap mean  $\pm$  bootstrap 95% CI, and drop  $s_{t,\text{PRE}} - s_{t,\text{POST}}$

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two are otherwise identical, so any difference reflects changes in the evaluation corpus rather than in the probe.

*Ablation* We flatten one prosodic feature at a time in the raw audio, re-summarise with the same model and prompt, and compare against the unaltered audio baseline; if the cue contributes measurably to quality, the ablated summary should degrade. Each manipulation isolates a single feature and leaves the other two intact. The energy is flattened by compressing the RMS envelope of active frames toward a constant target level while keeping silent frames unchanged. The pitch is flattened with the WORLD vocoder [8], setting voiced F0 to the per-file median while preserving the spectral envelope and aperiodicity; flattening to the per-file median rather than a fixed target avoids confounding the loss of variation with a shift in level. Pause is removed with the Silero voice-activity detector [17]. Non-speech regions are removed or compressed to a short residual gap, and the retained speech is concatenated in temporal order with equal-power cross-fades, preserving speech content and ordering.

### 3.3 Judge protocol

We complement the automatic metrics with an LLM-as-a-judge evaluation. Automatic metrics are insensitive to errors that matter most in minutes. A status inversion (a kept item reported as dropped) shares almost all of its tokens with the correct statement, but reverses its meaning, motivating an LLM-based evaluation. Two judges from different model families, Claude Opus 4.8 (Anthropic)<sup>6</sup>

<sup>6</sup> <https://www.anthropic.com/news/claude-opus-4-8>

and GPT-5.5 (OpenAI)<sup>7</sup> score each summary independently. Judges were selected from independent model families to minimise architecture-specific scoring biases. Each judge rates a summary against the human reference transcript on four metrics, each on a 1-5 Likert scale: *coverage* (are the important points present), *relevance* (is the content substantive and correctly placed), *factuality* (are the stated facts correct), and *groundedness* (is the content traceable to the transcript). Both judges receive the same scoring prompt, which defines each metric and its scale and requests a per-section score; corpus-adapted only in naming the third section, Actions on AMI and Progress on ICSI. For fair comparison, both judges remained blind to modality, with the two summaries of a meeting presented under arbitrary A/B labels and scored against the same transcript in a single pass, so a score difference reflects the summaries rather than the judge variance. The label assignment is fixed within each corpus with audio as A on AMI, text as A on ICSI, so label order is confounded with modality within a corpus but reversed between corpora. Although label order was reversed between corpora, residual order effects cannot be fully excluded and should be considered when interpreting small differences.

## 4 Results

### 4.1 Audio vs Text

*Automatic evaluation* Table 1 reports both corpora under both metrics, each mean with its bootstrap 95% CI. Because these per-condition intervals describe individual means rather than the paired gap, significance is assessed by the paired bootstrap of the mean audio–text difference (percentile,  $B=10,000$ ), reported below. The two metrics split cleanly into a semantic (BERTScore) and a lexical (ROUGE-Lsum) component, and the contrast between corpora is the main finding. Semantically on AMI, audio exceeds text in all three commitment-bearing sections and the paired difference excludes zero in each (Decisions  $+0.008$  [ $+0.004, +0.013$ ], Actions  $+0.013$  [ $+0.003, +0.021$ ], Problems  $+0.015$  [ $+0.011, +0.019$ ]), with Abstract at parity ( $+0.001$  [ $-0.002, +0.004$ ]). On ICSI the same comparison is null in every section (largest  $+0.006$  [ $-0.002, +0.012$ ]), so the semantic edge narrows to parity rather than reversing. Lexically the AMI pattern is section-dependent: audio leads on Problems ( $+0.035$  [ $+0.013, +0.057$ ]) while text reuses more exact gold wording in the Abstract ( $-0.024$  [ $-0.042, -0.008$ ]), reflecting more paraphrasing by the audio condition with preserved meaning; Decisions and Actions are indeterminate. On ICSI the clean transcript leads decisively, with text ahead in Decisions, Progress, and Problems (all CIs excluding zero) and winning the per-meeting ROUGE-Lsum mean on all six meetings (exact Wilcoxon  $p=.031$ ). The audio advantage is therefore narrow and corpus-specific.

*LLM-as-a-judge evaluation* The two judges agree substantially (quadratic-weighted  $\kappa = 0.64$ ), and Table 2 reports their per-section scores in both corpora. The dominant effect is coverage, where text scores above audio in every section on both

<sup>7</sup> <https://openai.com/index/introducing-gpt-5-5/>

**Table 1.** Audio-versus-text results: ROUGE-Lsum and BERTScore F1 by corpus, section, and condition, duration-matched at 40 min. AMI:  $n = 20/18/17/18$  by section; ICSI:  $n = 6/5/6/6$ . The third section is Actions (AMI) / Progress (ICSI). Each entry is the mean with its bootstrap 95% CI (percentile,  $B=10,000$ ) shown as a subscript; bold marks the better condition per section.

| Corpus | Metric / Condition | Abstract                     | Decisions                    | Actions/Progress             | Problems                     |
|--------|--------------------|------------------------------|------------------------------|------------------------------|------------------------------|
| AMI    | ROUGE-Lsum Audio   | 0.214<br>[.197, .234]        | 0.294<br>[.237, .355]        | <b>0.312</b><br>[.222, .411] | <b>0.229</b><br>[.195, .264] |
|        | ROUGE-Lsum Text    | <b>0.239</b><br>[.218, .259] | <b>0.312</b><br>[.258, .370] | 0.269<br>[.205, .336]        | 0.194<br>[.162, .229]        |
|        | BERTScore Audio    | <b>0.856</b><br>[.852, .861] | <b>0.870</b><br>[.863, .879] | <b>0.885</b><br>[.876, .895] | <b>0.873</b><br>[.869, .878] |
|        | BERTScore Text     | 0.856<br>[.852, .860]        | 0.862<br>[.854, .871]        | 0.873<br>[.864, .882]        | 0.858<br>[.854, .863]        |
| ICSI   | ROUGE-Lsum Audio   | <b>0.260</b><br>[.210, .304] | 0.227<br>[.188, .286]        | 0.219<br>[.183, .264]        | 0.220<br>[.174, .269]        |
|        | ROUGE-Lsum Text    | 0.242<br>[.222, .261]        | <b>0.276</b><br>[.238, .309] | <b>0.267</b><br>[.237, .305] | <b>0.264</b><br>[.202, .314] |
|        | BERTScore Audio    | <b>0.856</b><br>[.844, .869] | <b>0.856</b><br>[.849, .862] | <b>0.852</b><br>[.843, .861] | 0.852<br>[.842, .861]        |
|        | BERTScore Text     | 0.855<br>[.847, .864]        | 0.850<br>[.841, .860]        | 0.847<br>[.840, .853]        | <b>0.854</b><br>[.850, .856] |

corpora, plausibly in part because text summaries are longer and hence cover more reference content. Audio’s advantage is decision-level factuality on AMI, where both judges place audio above text (4.35/4.10 vs. 3.95/4.00); this does not generalise, as the same comparison is flat or reversed on ICSI. Relevance and groundedness are close throughout, with the text perfectly grounded in ICSI. The evaluation thus shows a coverage-fidelity trade-off as text recovers more content, while audio is marginally more faithful per the asserted decision on AMI.

## 4.2 Probing

We report probe recoverability at both taps. On AMI, energy is strongly recoverable before the adapter ( $R^2 = 0.71$ ) but collapses to non-recoverable after it, and pause survives only partially (AUROC  $0.96 \rightarrow 0.89$ ); pitch is not linearly recoverable at either tap ( $R^2 \leq 0$ ). The locus is the adapter’s 4:1 compression, which collapses the effective rank of the representation from 327 to 82. This reduction in effective rank suggests that adapter compression acts as an information bottleneck, preferentially preserving linguistic content while attenuating fine-grained acoustic variation. Evaluating the same AMI-trained probes on ICSI reproduces this pattern: energy is already weaker before the adapter ( $R^2 = 0.39$ ) and again drops to non-recoverable after it, pause attenuates ( $0.92 \rightarrow 0.82$ ), and pitch remains unrecoverable at both taps. Every PRE-to-POST drop is significant in both corpora (bootstrap 95% CIs exclude zero). The lower pre-adapter energy on ICSI (0.39 vs. 0.71 on AMI) is consistent with the domain shift to



**Table 2.** Per-judge mean LLM scores (1-5) for Audio (A) and Text (T) inputs, by metric and section, on AMI and ICSI. J1 = Claude Opus 4.8, J2 = GPT-5.5. Third section is Actions (AMI) / Progress (ICSI). Bold marks the higher score within each A/T pair; ties are unbolded.

| Metric       | Section    | AMI         |             |             |             | ICSI        |             |             |             |
|--------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|              |            | J1          |             | J2          |             | J1          |             | J2          |             |
|              |            | A           | T           | A           | T           | A           | T           | A           | T           |
| Coverage     | Abstract   | 3.90        | <b>4.70</b> | 4.10        | <b>4.95</b> | 2.50        | <b>3.50</b> | 3.50        | <b>4.17</b> |
|              | Decisions  | 3.55        | <b>4.05</b> | 3.70        | <b>4.40</b> | 2.33        | <b>3.50</b> | 2.83        | <b>4.33</b> |
|              | Act./Prog. | 3.10        | <b>3.90</b> | 3.55        | <b>4.45</b> | 2.83        | <b>3.17</b> | 3.50        | <b>4.33</b> |
|              | Problems   | 3.45        | <b>4.05</b> | 3.85        | <b>4.60</b> | 2.67        | <b>3.50</b> | 3.83        | <b>4.00</b> |
| Relevance    | Abstract   | <b>5.00</b> | 4.95        | 4.95        | <b>5.00</b> | 4.33        | <b>4.67</b> | 4.50        | <b>4.67</b> |
|              | Decisions  | <b>4.65</b> | 4.55        | <b>4.55</b> | 4.35        | 4.33        | <b>4.50</b> | 4.17        | <b>4.83</b> |
|              | Act./Prog. | 3.55        | <b>4.10</b> | 3.45        | <b>3.85</b> | <b>5.00</b> | 4.67        | 4.67        | <b>5.00</b> |
|              | Problems   | 3.70        | 3.70        | 4.15        | <b>4.65</b> | 4.00        | <b>4.33</b> | 4.50        | <b>4.67</b> |
| Groundedness | Abstract   | 5.00        | 5.00        | 5.00        | 5.00        | 4.83        | <b>5.00</b> | 5.00        | 5.00        |
|              | Decisions  | <b>4.90</b> | 4.55        | <b>4.85</b> | 4.70        | 4.67        | <b>5.00</b> | 4.83        | <b>5.00</b> |
|              | Act./Prog. | 4.45        | <b>4.60</b> | 4.40        | 4.40        | 5.00        | 5.00        | 4.83        | <b>5.00</b> |
|              | Problems   | 4.25        | <b>4.70</b> | 4.55        | <b>4.80</b> | 4.67        | <b>5.00</b> | 5.00        | 5.00        |
| Factuality   | Abstract   | 4.90        | <b>5.00</b> | 4.85        | <b>4.95</b> | <b>4.83</b> | 4.67        | <b>4.67</b> | 4.50        |
|              | Decisions  | <b>4.35</b> | 3.95        | <b>4.10</b> | 4.00        | <b>3.67</b> | 3.33        | 3.50        | <b>4.00</b> |
|              | Act./Prog. | <b>3.60</b> | 3.55        | 3.50        | <b>3.60</b> | <b>4.50</b> | 4.00        | 4.00        | <b>4.33</b> |
|              | Problems   | 3.35        | <b>3.75</b> | 3.90        | <b>4.25</b> | 3.50        | <b>4.67</b> | <b>4.33</b> | 4.33        |

research-meeting speech and the change of channel; the quantity that transfers is the attenuation across the adapter, not its level. The bottleneck is therefore a property of the model rather than of the AMI dataset.

### 4.3 Ablation

Table 3 reports per-section, per-feature deltas (ablated minus baseline) on both corpora. The result is a clean null. In BERTScore, every ablation stays within 0.007 of baseline in both AMI and ICSI, and in ROUGE-Lsum the deltas are small and unsystematic. The few intervals that exclude zero are small and predominantly positive; that is, ablation slightly *raises* the score, which is the opposite of what a feature carrying useful information would produce. Removing pitch, energy, or pause, therefore, leaves summary quality unchanged, consistent with these cues being redundant with the words in this setting.

Three checks distinguish this null from an underpowered one. First, equivalence testing: in BERTScore the AMI manipulations are statistically equivalent to no effect within a 0.01 margin (TOST<sup>8</sup>,  $p < 0.05$ , all sections), while ROUGE-Lsum is reported as no significant change rather than equivalence, since at  $n \approx 20$  the test is underpowered at a defensible margin. Second, manipulation strength: each ablation was independently verified to be strong before any null was trusted,

<sup>8</sup> Two One-Sided Tests

**Table 3.** Ablation deltas (ablated – baseline) for ROUGE-Lsum ( $\Delta R$ ) and BERTScore ( $\Delta B$ ), on AMI ( $n=20$ ) and ICSI ( $n=6$ ). \* marks a paired bootstrap 95% CI excluding zero. Ab=Abstract, De=Decisions, AP=Actions/Progress, Pr=Problems.

| Feature | Metric     | AMI    |        |       |       | ICSI   |        |        |        |
|---------|------------|--------|--------|-------|-------|--------|--------|--------|--------|
|         |            | Ab     | De     | AP    | Pr    | Ab     | De     | AP     | Pr     |
| –pitch  | $\Delta R$ | +.012* | +.007  | –.016 | –.006 | +.020  | +.031* | +.023  | +.022  |
|         | $\Delta B$ | .000   | –.003  | –.002 | +.001 | +.007* | +.001  | –.002  | +.004* |
| –energy | $\Delta R$ | +.016  | +.019* | –.004 | –.005 | –.008* | +.026  | –.019  | –.007  |
|         | $\Delta B$ | –.001  | –.001  | +.001 | –.001 | +.001  | +.001  | –.004* | +.002  |
| –pause  | $\Delta R$ | +.012  | +.019  | –.012 | –.016 | –.012  | +.021* | +.000  | +.019  |
|         | $\Delta B$ | .000   | .000   | +.001 | –.001 | .000   | +.003  | .000   | +.005  |

for example the F0 spread collapsing from  $\approx 21$  Hz to  $\approx 2.4$  Hz. For pitch, the flattened audio additionally passes through WORLD resynthesis, so the comparison is conservative, as any vocoder degradation would push the ablated summary toward lower quality, yet none is observed. Third, a positive control: the manipulations altered the generated text in 79 of 80 AMI meeting-sections without changing its quality, confirming that the model responds to the perturbation but does not use prosody to improve correctness. The overall pattern suggests that audio primarily affects how information is expressed rather than what information is captured.

## 5 Discussion

*Interpretation* On **RQ1**, audio and text serve different priorities rather than one dominating the other: the audio model earns its compute where decision-level fidelity outweighs coverage, an edge that holds on AMI but weakens on naturalistic ICSI speech. The advantage is therefore real but conditional, not a general property of the audio channel. On **RQ2** and **RQ3**, the three tested cues fail both halves of our criterion. Probing locates the failure of representation at the adapter, whose compression removes the energy and pause detail recoverable at the encoder while pitch is never linearly recoverable; ablation locates the failure of necessity, since flattening any cue leaves quality unchanged. Together these explain why prosody does not account for the advantage: the surviving detail is redundant with the lexical channel once a strong decoder is in place. This leaves the AMI edge unexplained by prosody. Pitch, energy, and pause are only part of the audio signal: speaker identity, overlap and turn-taking, and voice quality also survive in audio but not the transcript, and none is captured by our three features. The residual advantage plausibly lives there, untested here.

*Limitations* Four boundaries scope our claims. First, the mechanistic findings are based on a single architecture, Voxtral-Small-24B; the bottleneck is a property of this adapter design, and whether it generalises to other adapters remains untested. No adapter-free speech LLM exists at this scale, so our claim is

scoped to adapter-based architectures. Second, the 40-minute cap leaves longer meetings unexamined, where lexical redundancy may weaken and prosody could matter more; all six ICSI meetings were truncated, so the scored span may omit later decisions and understate any audio advantage. Third, both corpora are English; prosody carries language-dependent functions in tonal and pitch-accent languages, so our redundancy finding may not hold cross-linguistically. Finally, our probes are linear and our ablations test necessity rather than sufficiency. Consequently, nonlinear probes may recover representational structure that is not detectable with our approach, and our conclusions regarding representation should therefore be interpreted within the limits of probe capacity.

*Ethics* AMI and ICSI are public, consent-based corpora, so no private data was collected here. The LLM judges are versioned commercial systems whose scores may carry model-specific bias and may not reproduce exactly across versions; we use two independent judges and a modality-blind protocol to limit this. Automatic minutes can contain factual errors, so they should serve as drafts for human review rather than authoritative records.

## 6 Conclusion

We asked how far audio improves meeting summarisation and what limits it. The answer is that the advantage is real but narrow: under a duration-matched comparison audio yields a semantic benefit on AMI that does not generalise to ICSI, and a modality-blind two-judge evaluation shows transcript summaries consistently cover more content, leaving fidelity to asserted decisions as audio’s only durable edge. Two mechanisms cap the advantage: prosodic detail is attenuated at the multimodal adapter, and what survives is redundant with the lexical channel once a strong decoder is in place. The cues we tested, pitch, energy, and pause, do not explain the AMI edge, whose source must lie elsewhere in the audio signal. We see two directions for future work: identifying which non-prosodic acoustic cues direct speech processing preserves, and testing whether adapters that retain more acoustic detail widen the narrow advantage observed here. More broadly, the findings suggest that future gains in speech-native LLMs may depend less on preserving traditional prosodic cues and more on retaining richer interactional and discourse-level information from the speech signal.

## References

1. Abouelenin, A., Ashfaq, A., Atkinson, A., Awadalla, H., Bach, N., Bao, J., Benhaim, A., Cai, M., Chaudhary, V., Chen, C., et al.: Phi-4-mini technical report: Compact yet powerful multimodal language models via mixture-of-loras. arXiv preprint arXiv:2503.01743 (2025)
2. Chu, Y., Xu, J., Yang, Q., Wei, H., Wei, X., Guo, Z., Leng, Y., Lv, Y., He, J., Lin, J., et al.: Qwen2-audio technical report. arXiv preprint arXiv:2407.10759 (2024)
3. Hellbernd, N., Sammler, D.: Prosody conveys speaker’s intentions: Acoustic cues for speech act perception. *Journal of Memory and Language* **88** (2016) 70–86

4. Kang, W., Roy, D.: Prompting large language models with audio for general-purpose speech summarization. In: Proc. Interspeech. (2024)
5. Lai, C., Renals, S.: Incorporating lexical and prosodic information at different levels for meeting summarization. In: Interspeech 2014. (2014) 1875–1879
6. Liu, A.H., Ehrenberg, A., Lo, A., Denoix, C., Barreau, C., Lample, G., Delignon, J.M., Chandu, K.R., von Platen, P., Muddireddy, P.R., et al.: Voxtral. arXiv preprint arXiv:2507.13264 (2025)
7. Lu, Y.J., Gao, K., Liang, M., Wang, H., Thebaud, T., Moro-Velazquez, L., Dehak, N., Villalba, J.: Spoken dialogsum: An emotion-rich conversational dataset for spoken dialogue summarization. arXiv preprint arXiv:2512.14687 (2025)
8. MORISE, M., YOKOMORI, F., OZAWA, K.: World: A vocoder-based high-quality speech synthesis system for real-time applications. IEICE Transactions on Information and Systems **E99.D**(7) (2016) 1877–1884
9. Murray, G., Renals, S., Carletta, J.: Extractive summarization of meeting recordings. In: Interspeech 2005. (2005) 593–596
10. Nielsen, E.: Effects of Prosody on Natural Language Processing. Phd thesis, University of Edinburgh (May 2024)
11. Nihei, F., Nakano, Y.I., Takase, Y.: Fusing verbal and nonverbal information for extractive meeting summarization. In: Proceedings of the Group Interaction Frontiers in Technology. GIFT’18, New York, NY, USA, Association for Computing Machinery (2018)
12. Qian, K., Fan, X., Ni, J., Shechtman, S., Hasegawa-Johnson, M.A., Gan, C., Zhang, Y.: ProsodyLM: Uncovering the emerging prosody processing capabilities in speech language models. In: Second Conference on Language Modeling. (2025)
13. Rennard, V., Shang, G., Hunter, J., Vazirgiannis, M.: Abstractive meeting summarization: A survey. Transactions of the Association for Computational Linguistics **11** (2023) 861–884
14. Retkowski, F., Züfle, M., Sudmann, A., Pfau, D., Watanabe, S., Niehues, J., Waibel, A.: Summarizing speech: A comprehensive survey. In Christodoulopoulos, C., Chakraborty, T., Rose, C., Peng, V., eds.: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, Suzhou, China, Association for Computational Linguistics (November 2025) 27275–27306
15. Riedhammer, K., Gillick, D., Favre, B., Hakkani-Tür, D.: Packing the meeting summarization knapsack. In: INTERSPEECH. (2008) 2434–2437
16. Tang, C., Yu, W., Sun, G., Chen, X., Tan, T., Li, W., Lu, L., MA, Z., Zhang, C.: Salmonn: Towards generic hearing abilities for large language models. In: The Twelfth International Conference on Learning Representations. (2024)
17. Team, S.: Silero vad: pre-trained enterprise-grade voice activity detector (vad), number detector and language classifier. <https://github.com/snakers4/silero-vad> (2024)
18. Wang, J.: ESSumm: Extractive Speech Summarization from Untranscribed Meeting. In: Proc. Interspeech 2022. (2022) 3243–3247
19. Wolf, L., Pimentel, T., Fedorenko, E., Cotterell, R., Warstadt, A., Wilcox, E.G., Regev, T.I.: Quantifying the redundancy between prosody and text. In Bouamor, H., Pino, J., Bali, K., eds.: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, Singapore, Association for Computational Linguistics (December 2023) 9765–9784
20. Xie, S., Hakkani-Tür, D., Favre, B., Liu, Y.: Integrating prosodic features in extractive meeting summarization. In: 2009 IEEE Workshop on Automatic Speech Recognition Understanding. (2009) 387–391